nonstatBaseball

2026-02-17

Simple Data

For this document I have a dataset with more rows, with larger amounts of Plate Appearances on average for each player. It uses more simple data that does not rely on Statcast metrics like Exit Velocity of Prospect Score.

```
head(base)
```

```
##   X.           Name  Team   G  PA HR  R RBI SB   BB.    K.   ISO BABIP   AVG
## 1 10     Alan Trejo   COL 188 512  9 53  48  6 0.053 0.232 0.101 0.277 0.224
## 2 11    Ji Hwan Bae   PIT 163 514  2 74  44 37 0.084 0.255 0.069 0.305 0.223
## 3 12 Carter Kieboom 2 Tms 136 516 12 57  43  0 0.101 0.269 0.100 0.258 0.200
## 4 13  Jon Singleton 2 Tms 158 519 15 57  54  0 0.116 0.260 0.138 0.272 0.218
## 5 14  Bryan Holaday 6 Tms 195 520  7 44  49  0 0.060 0.198 0.097 0.274 0.228
## 6 15    Matt Szczur 2 Tms 283 521  9 69  48  5 0.106 0.205 0.125 0.285 0.233
##     OBP   SLG  wOBA xwOBA wRC.  BsR   Off   Def  WAR Team_1            Level
## 1 0.269 0.325 0.261 0.271   49 -1.0 -32.9   4.2 -1.1    COL           A+,AA,R
## 2 0.294 0.293 0.265 0.270   62  4.6 -19.2  -5.5 -0.7    PIT              A,R
## 3 0.297 0.300 0.269 0.282   64 -1.9 -25.3 -12.5 -2.1    WSN A,A-,A+,AA,AAA,R
## 4 0.306 0.356 0.294 0.312   91 -4.5 -10.6 -14.5 -0.8    PHI      A,A+,AA,AAA
## 5 0.282 0.325 0.263 0.248   61 -3.8 -28.4   3.8 -0.8    DET        A+,AA,AAA
## 6 0.322 0.358 0.300 0.279   84 -2.7 -12.8  -0.2  0.4    CHC A,A-,A+,AA,AAA,R
##     Age PA_1      BB._1      K._1    BB.K_1     AVG_1     OBP_1     SLG_1
## 1 21-23 1163 0.05588994 0.2115219 0.2642276 0.2747875 0.3228895 0.4343720
## 2 18-19  532 0.10902256 0.1748120 0.6236559 0.3085339 0.3913043 0.4070022
## 3 18-21 1462 0.11627907 0.2010944 0.5782313 0.2867355 0.3775650 0.4694202
## 4 18-25 3590 0.16016713 0.2353760 0.6804734 0.2549020 0.3752089 0.4526707
## 5 22-31 1843 0.07596310 0.1785133 0.4255319 0.2481527 0.3186813 0.3713054
## 6 20-29 2671 0.08386372 0.1433920 0.5848564 0.2838520 0.3491884 0.3994954
##       OPS_1      ISO_1      Spd   BABIP_1      wSB_1     wRC_1     wRAA_1
## 1 0.7572615 0.15958451 4.102931 0.3274336 -3.2497689 153.94440   9.156977
## 2 0.7983065 0.09846827 8.161438 0.3810811  1.9898806  83.48946  20.644794
## 3 0.8469852 0.18268467 4.803504 0.3390558  0.2119062 241.83108  55.658573
## 4 0.8278796 0.19776876 3.402095 0.3075780 -1.9085170 551.34607 113.309218
## 5 0.6899867 0.12315271 2.616653 0.2882812 -1.7865396 197.35962 -17.010717
## 6 0.7486837 0.11564340 6.909374 0.3234107  7.5704149 343.97373  23.136109
##      wOBA_1    wRC._1 PlayerId
## 1 0.3365582  96.00167    20056
## 2 0.3717883 136.24032    23818
## 3 0.3795992 130.41936    19958
## 4 0.3704578 126.50592    10441
## 5 0.3145318  92.56852    11287
## 6 0.3420112 106.08557    11255
```

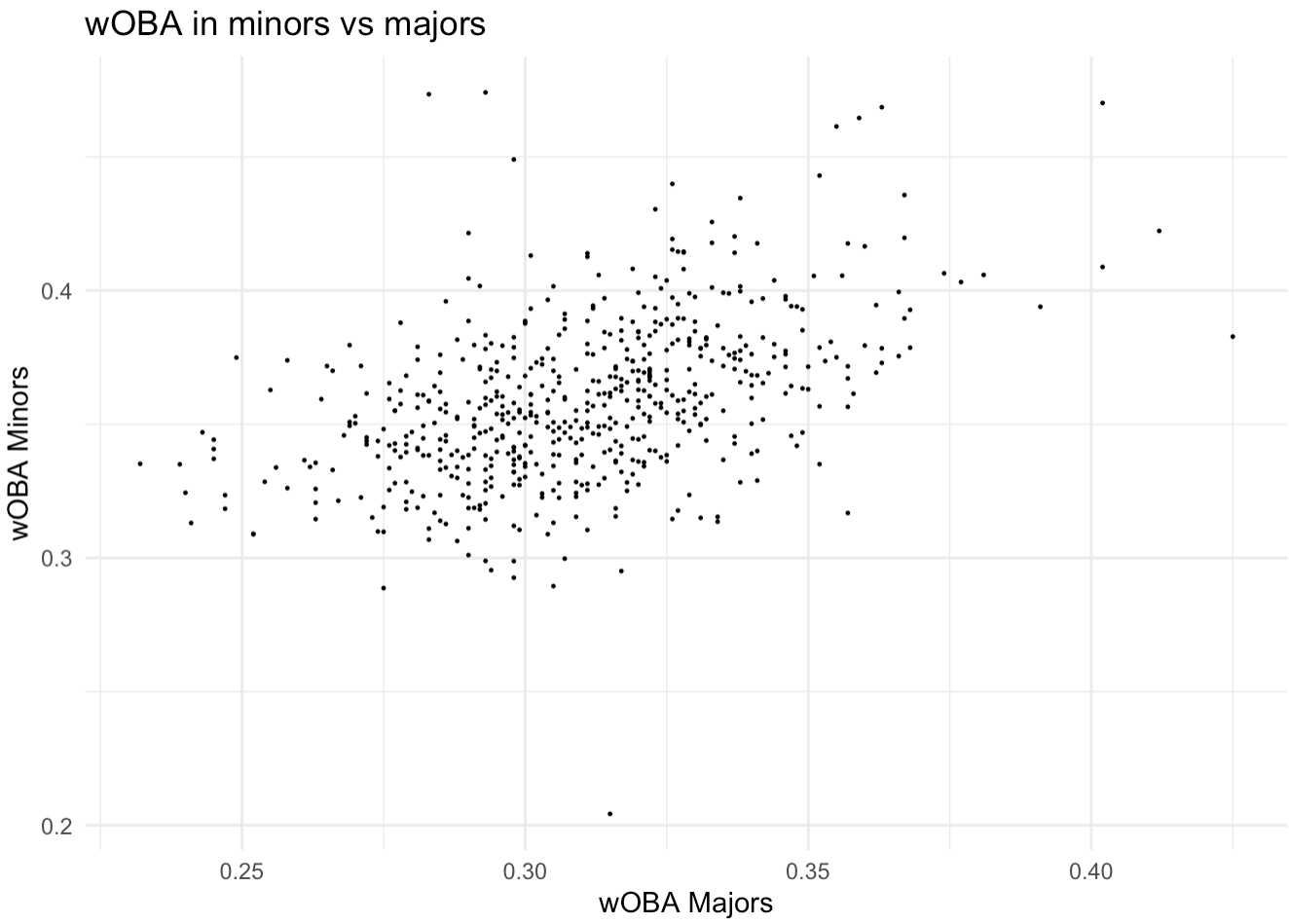
```
nrow(base)
```

```
## [1] 591
```

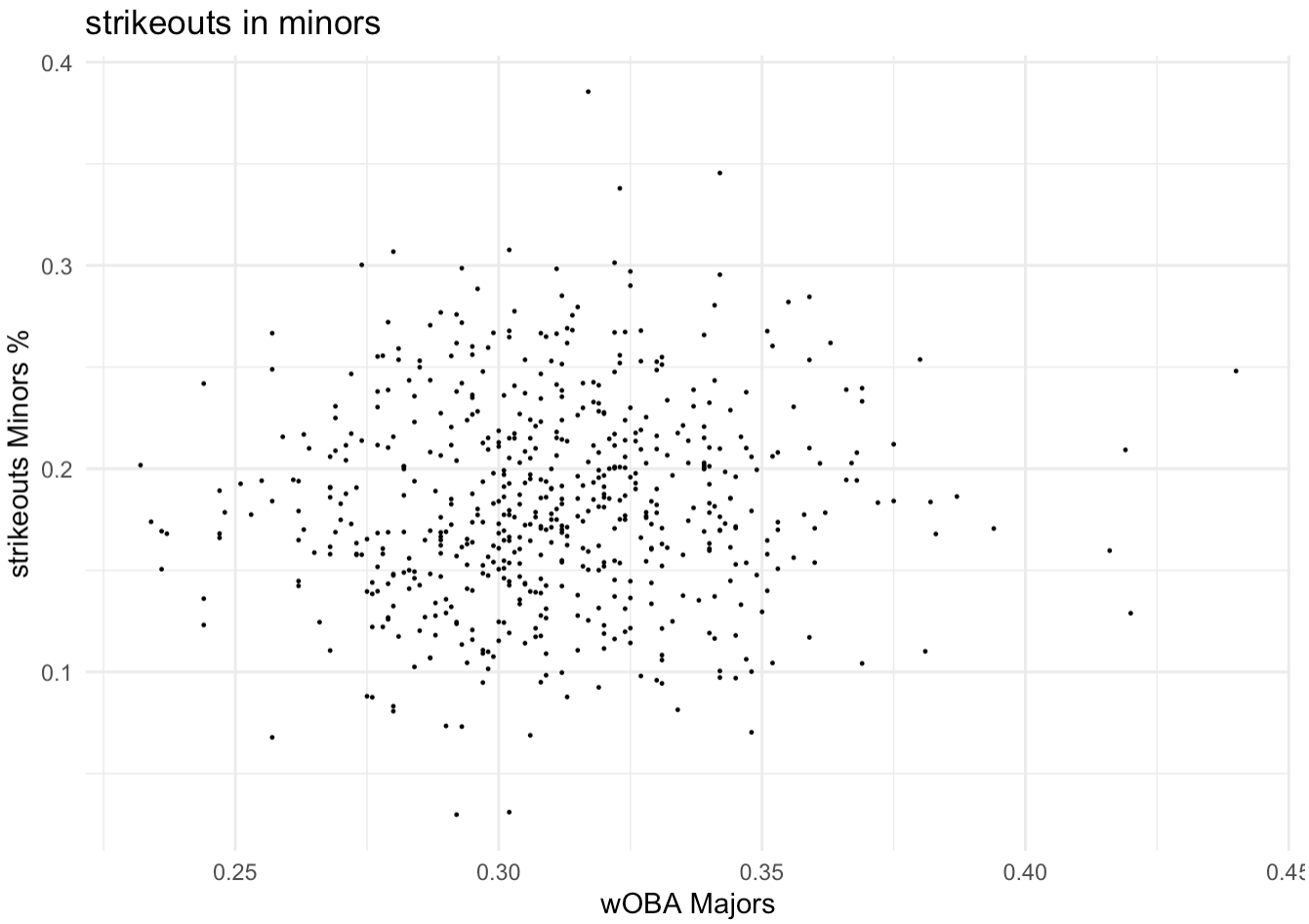
We can see here we have a similar number of columns, but we can see we have four times as many rows, and an increase in Plate Appearances that are even more important to reducing variance.

Graphs

```
ggplot(base, aes(wOBA, wOBA_1)) +
  geom_point(color="black", size = 0.2) +
  labs(title = "wOBA in minors vs majors",
       x = "wOBA Majors",
       y = "wOBA Minors") +
  theme_minimal()
```



```
ggplot(base, aes(xwOBA, K._1)) +
  geom_point(color="black", size = 0.2) +
  labs(title = "strikeouts in minors",
       x = "wOBA Majors",
       y = "strikeouts Minors %") +
  theme_minimal()
```



We can see in this graph that there is a much stronger corrolation with pure hitting stats then strikeouts. wOBA is a stat that quantifies all parts of hitting, and gives a highers score to hitters that hit better. We can see that if you hit better in the minors then you hit better in the majors, but it is a fairly loose correlation. With strikeout percentage, there is an extremely small correlation so, if you strikeout less, you have a small chance of preforming better in the majors.

```
wOBA <- lm(wOBA_1 ~ wOBA, base)
summary(wOBA)
```

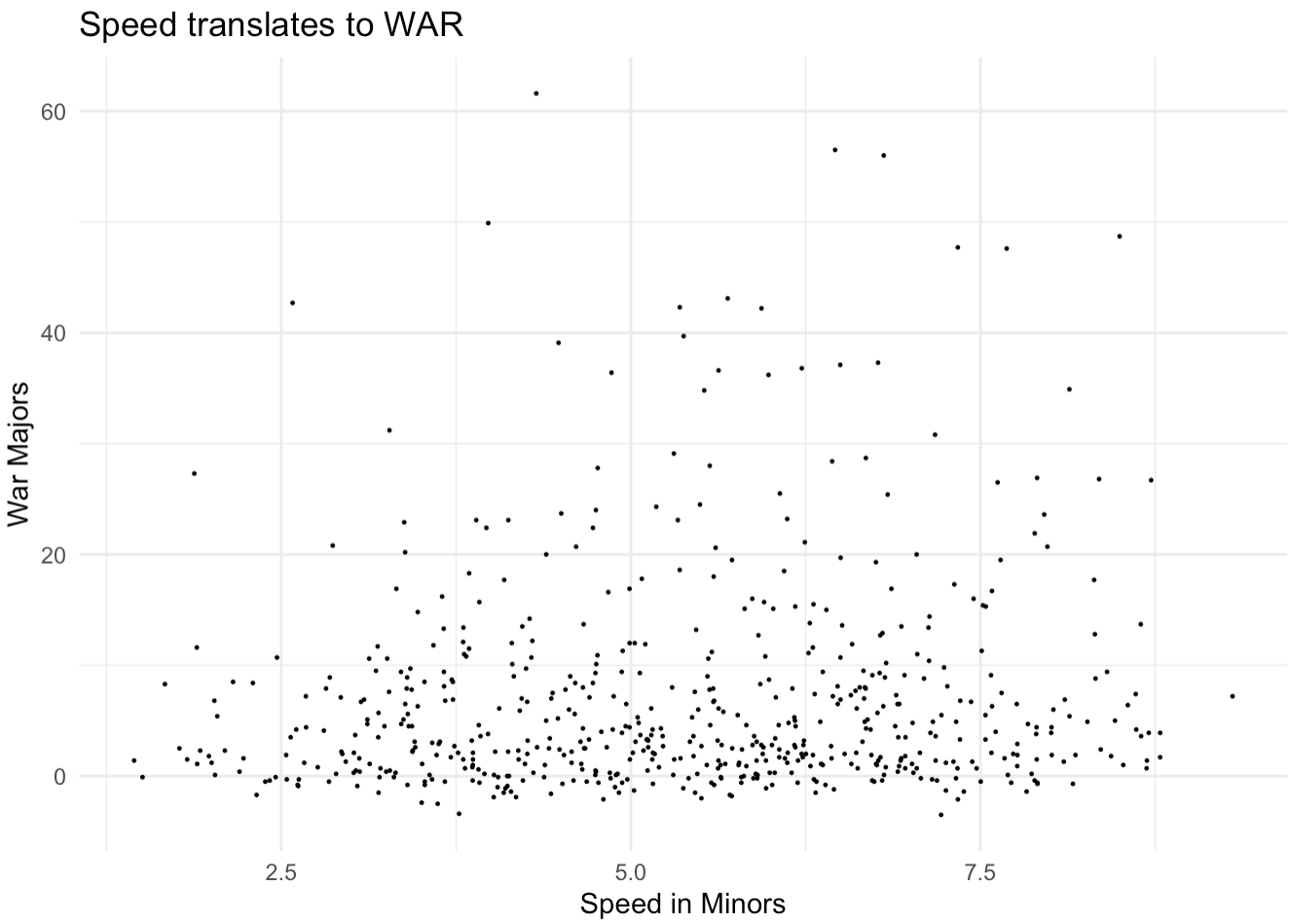
```
##
## Call:
## lm(formula = wOBA_1 ~ wOBA, data = base)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.15714 -0.01589 -0.00103  0.01525  0.12879
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.19670    0.01266   15.54   <2e-16 ***
## wOBA         0.52289    0.04066   12.86   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02687 on 589 degrees of freedom
## Multiple R-squared:  0.2192, Adjusted R-squared:  0.2179
## F-statistic: 165.4 on 1 and 589 DF,  p-value: < 2.2e-16
```

```
k <- lm(K._1 ~ wOBA, base)
summary(k)
```

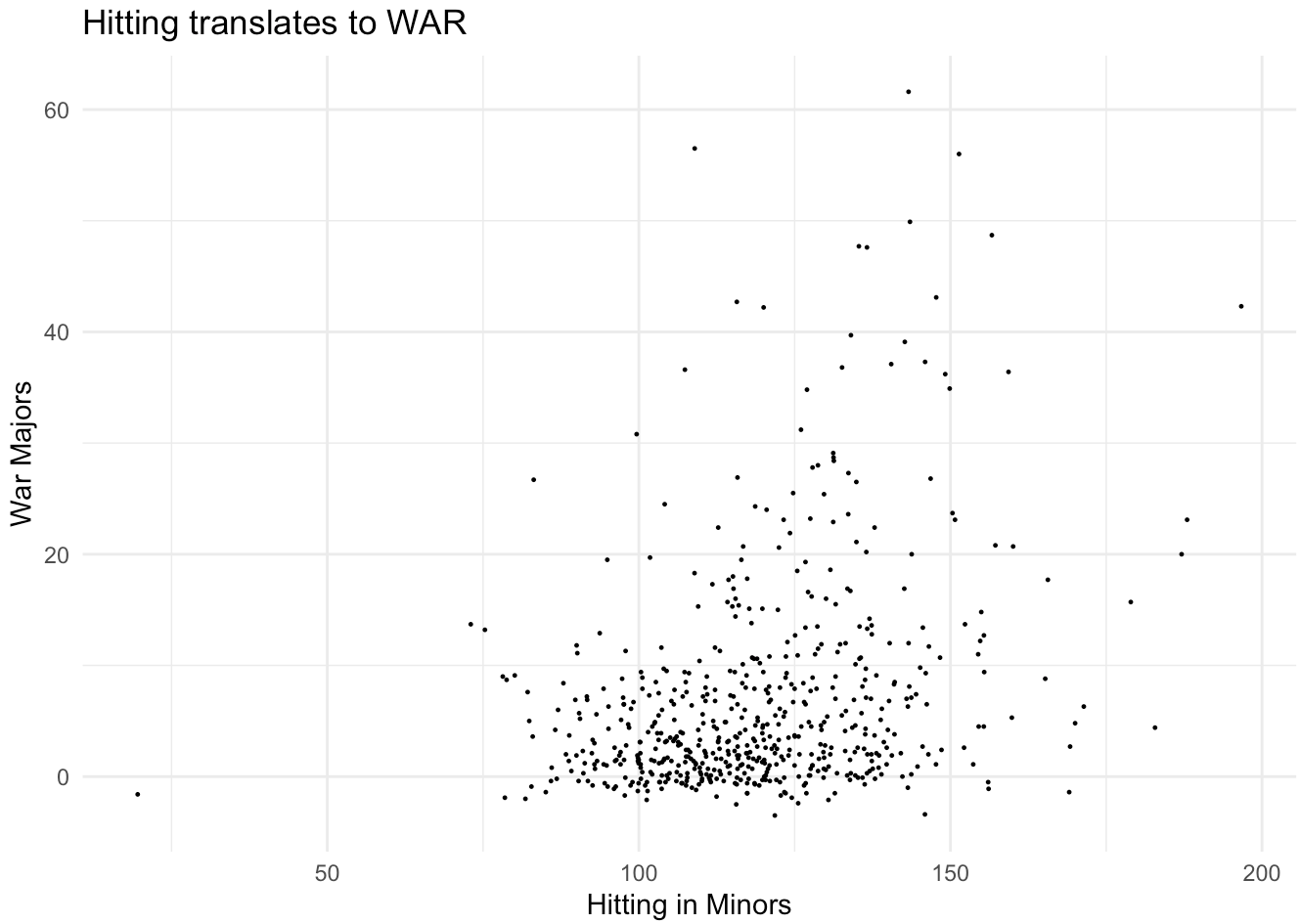
```
##
## Call:
## lm(formula = K._1 ~ wOBA, data = base)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.153118 -0.032246 -0.003411  0.031270  0.201977
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.17411    0.02366   7.359 6.26e-13 ***
## wOBA         0.02973    0.07600   0.391    0.696
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05023 on 589 degrees of freedom
## Multiple R-squared:  0.0002597,  Adjusted R-squared:  -0.001438
## F-statistic: 0.153 on 1 and 589 DF,  p-value: 0.6958
```

We can see that there is no significance for strikeouts, with the p-value being well below alpha level 0.01, and there being absolutely no R^2. However, there is significance for the wOBA hitting stat, with a small R^2, showing while it is a weak relation, hitting in the minors is much more important for a prospects success.

```
ggplot(base, aes(Spd, WAR)) +
  geom_point(color="black", size = 0.2) +
  labs(title = "Speed translates to WAR",
       x = "Speed in Minors",
       y = "War Majors") +
  theme_minimal()
```




```
ggplot(base, aes(wRC._1, WAR)) +
  geom_point(color="black", size = 0.2) +
  labs(title = "Hitting translates to WAR",
       x = "Hitting in Minors",
       y = "War Majors") +
  theme_minimal()
```



We can see pure athleticism quantified by the Spd stat that measures you running speed in the minors, but we cannot see a visible effect on your WAR in the Majors. This means that being fast is not a good way to project a good baseball player. Even hitting in the minors well, does not mean you are necessarily going to do well in the majors, as there is a ton of variance.

```
wrc <- lm(wRC._1 ~ WAR, base)
summary(wrc)
```

```
##
## Call:
## lm(formula = wRC._1 ~ WAR, data = base)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -94.208 -11.234  -0.886  11.281  65.430
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 114.74305    0.92267 124.360  < 2e-16 ***
## WAR           0.60447    0.07675   7.876 1.64e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.16 on 589 degrees of freedom
## Multiple R-squared:  0.09528,    Adjusted R-squared:  0.09374
## F-statistic: 62.03 on 1 and 589 DF,  p-value: 1.638e-14
```

```
Spd <- lm(WAR ~ Spd + wRC._1, base)
summary(Spd)
```

```
##
## Call:
## lm(formula = WAR ~ Spd + wRC._1, data = base)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -17.736  -5.703  -2.369   2.788  51.330
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -14.72882    2.67880  -5.498 5.72e-08 ***
## Spd           0.57465    0.22563   2.547   0.0111 *
## wRC._1        0.15714    0.01992   7.888 1.51e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.231 on 588 degrees of freedom
## Multiple R-squared:  0.1051, Adjusted R-squared:  0.1021
## F-statistic: 34.55 on 2 and 588 DF,  p-value: 6.527e-15
```

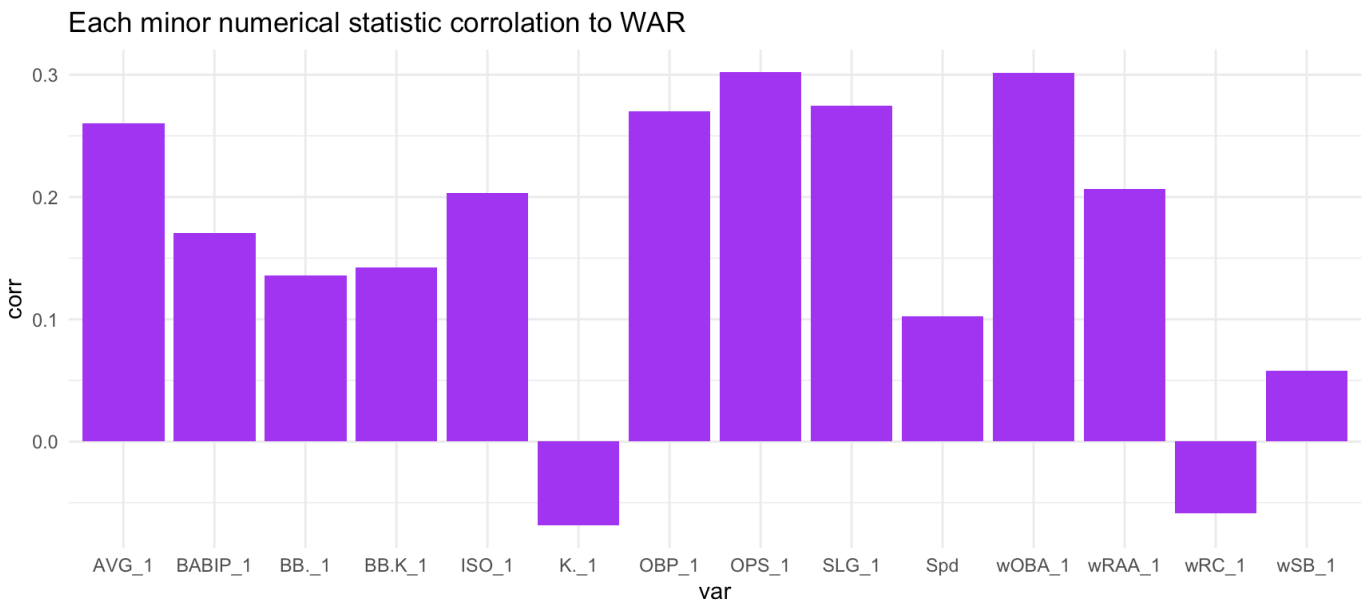
We can also see using linear regression, that there is a significant relation between Hitting in the Minors and your WAR in the majors. However, Spd is just not significant, however the R^2 for both are very small, which lines up with what we see in the graphs, with there being very little cause and effect.

Correlation

```
minn <- base[, 28:41]

fit <- data.frame(
  var = colnames(minn),
  corr = NA
)

for (x in 1:ncol(minn)) {
  fit$corr[x] <- cor(base[, 23], minn[, x])
}

fit
```

```
##        var        corr
## 1    BB._1  0.13583250
## 2     K._1 -0.06829285
## 3   BB.K_1  0.14217347
## 4    AVG_1  0.26049398
## 5    OBP_1  0.27020507
## 6    SLG_1  0.27446395
## 7    OPS_1  0.30208816
## 8    ISO_1  0.20354626
## 9      Spd  0.10231245
## 10 BABIP_1  0.17055927
## 11   wSB_1  0.05818559
## 12   wRC_1 -0.05856860
## 13  wRAA_1  0.20685346
## 14  wOBA_1  0.30143691
```

```
ggplot(fit, aes(x = var, y = corr)) +
  geom_col(fill = "purple") +
  labs(title = "Each minor numerical statistic corrolation to WAR") +
  theme_minimal()
```

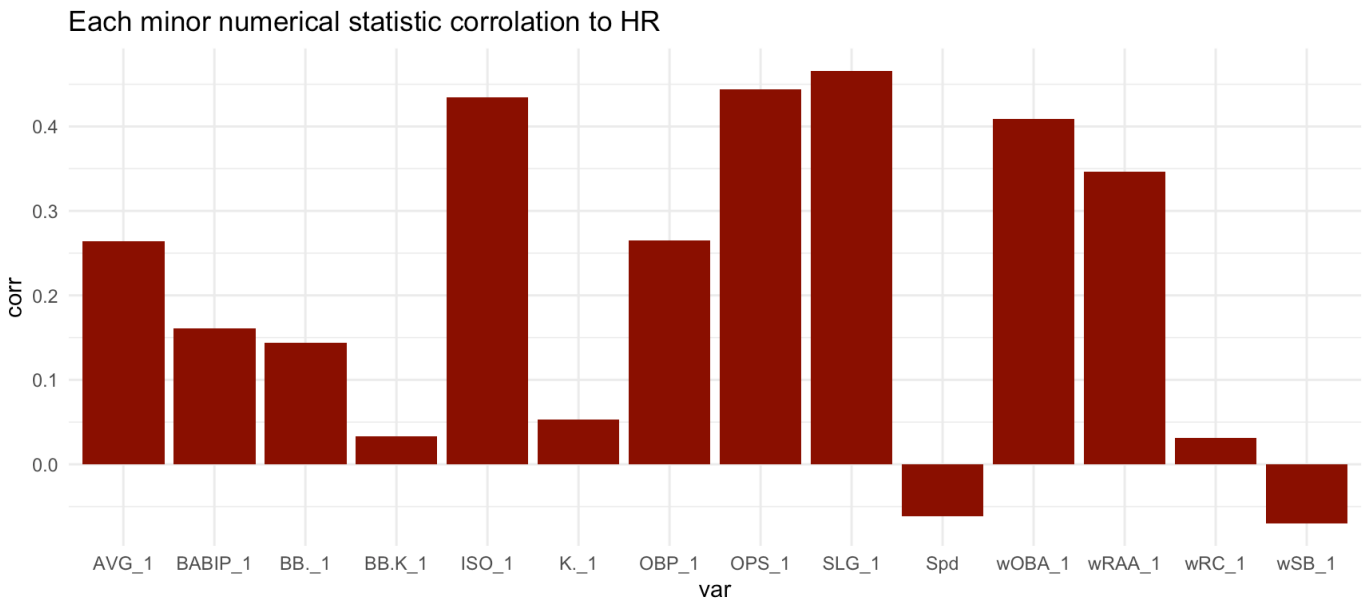


We can see here that most have positive relationships, with the big exception being strikeout percentage, which makes sense, as a higher strikeout percentage means you are hitting the ball less. We can see the strongest correlation comes from Batting Average, and wOBA or OPS, which is interesting. wOBA and OPS supposed to be a precise stat that fully quantifies your hitting ability, while batting average is just a part of those statistics, but it is nearly as strong.

```
fit2 <- data.frame(
  var = colnames(minn),
  corr = NA
)

for (x in 1:ncol(minn)) {
  fit2$corr[x] <- cor(base[, 6], minn[, x])
}

fit2
```

```
##        var        corr
## 1    BB._1  0.14426239
## 2     K._1  0.05287082
## 3   BB.K_1  0.03357407
## 4    AVG_1  0.26373744
## 5    OBP_1  0.26553067
## 6    SLG_1  0.46562828
## 7    OPS_1  0.44395081
## 8    ISO_1  0.43487240
## 9      Spd -0.06170955
## 10 BABIP_1  0.16071642
## 11   wSB_1 -0.06978059
## 12   wRC_1  0.03162089
## 13  wRAA_1  0.34661870
## 14  wOBA_1  0.40914995
```

```
ggplot(fit2, aes(x = var, y = corr)) +
  geom_col(fill = "darkred") +
  labs(title = "Each minor numerical statistic corrolation to HR") +
  theme_minimal()
```



Here we look at the most important stats that lead to home runs, as that is something that is very important in the modern day, we can see Speed is one of the few negative values, which makes sense given power hitters are usually not speedy. We can see the best stats are the ones focused on power like Isolated Power, and Slugging. We can also see that there is little correlation between strikeouts and Home Runs, which shows that you don't necessarily strikeout more if you are going for Home Runs, however we see Walk rate and On Base are highly correlated, which shows that having a good eye for balls is good for home runs.